

## Session 6: CPSA Goals

Slides prepared by UMBC Protocol Analysis Lab

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## Common Security Goals

- ▶ **Confidentiality:** In a two-party communication protocol, confidentiality holds if no third party is able to access the information sent.
- ▶ **Agreement:** In a two-party communication protocol, agreement holds if both parties agree on some set of data values in the protocol. These values are then used to derive the key.

## Other common security goals

1. Aliveness
2. Weak Agreement
3. Non-injective Agreement
4. A specific relay party cannot read the message
5. The message must be timestamped
6. Perfect forward secrecy
7. Crypto primitives are resistant to a predefined attack



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Confidentiality and Agreement (on the secret) needed.

# CPSA Goals

- ▶ Protocol goals as mathematical statements
- ▶ Find strands that satisfy (or don't satisfy) some set of properties
- ▶ May reduce time taken to terminate

# Some more tips for reading specifications

- ▶ FIND THE SECURITY GOALS
  - ▶ Are additional SGs needed?
- ▶ Modularize the spec into blocks of messages
  - ▶ “In the first set of exchanges”
  - ▶ “[party1] preshares [data] with [party2]”
- ▶ Ignore nonsense
  - ▶ “[protocol] does not consider [niche usecase] because. . .”
  - ▶ “We assume clients have access to an ALU” (structural vs. implementation)
  - ▶ “Our conformance program handles...”

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- ▶ Let's try modeling it!

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# Survey

Fill in details about the survey here!